

Introduction to Arduino and microcontrollers

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- A long time ago, working on an electronic circuitry to do a specific job meant building a complex electronic design of components such as resistors, capacitors, files, transistors, etc.

- The electronic circuits were fixed in design and a change or modification of a simple part in it meant a lot of complex processes such as welding and cutting wires and re-examination of electronic drawings and many of the annoying things that led to limiting the function of developing electronic products to a group of specialized engineers only.

- With the advent of semiconductor technology and the invention of Integrated Circuits (IC), it has become possible to develop a complete electronic circuit on a small chip, which may not exceed the pinhead, so that at present there are electronic circuits of a size that is estimated in the nanometer, which can not be seen Except using special light amplifiers.

- The development of Integrated Circuits has led to the emergence of a special generation of electronic circuits called Microcontrollers, which is more like a programmable microcomputer to perform a range of functions such as reading temperature, controlling electric motor, or even managing production lines in large factories.
- Through the orders of programming and thus transformed the technology of the manufacture of electronic circuits and systems from the pure electronic design based on the solid components only to the orders of the software can be individually written and designed by himself easily .

- Microcontrollers also have the ability to change and modify at any time. If you want to change something in your project, you can modify it in the programming lines and re - place new commands on the Microcontroller and experiment more than once until you reach your project for the desired goal.

- Arduino is an open-source electronic board for the development of a lot of ideas and projects related to control in a simple way by using an open-source programming language Arduino C
- The microcontroller is located on the board and programmed using Arduino IDE: Integrated Development Environment, which is free to download.

- What distinguishes Arduino boards is the ease of handling and simplicity of the programming language, which has been working by a team from Italy since 2005 until now.
- The Arduino language has been derived from the language “processing” and the language "C "Which is the basis of modern programming languages .

- Many people believe that Arduino is designed for amateurs only, but this is not true, it has been developed to suit all levels of amateur to advanced projects and evidence that there are great advantages that make Arduino the top microcontroller is the possibility of integration in projects programmed in advanced engineering languages such as MATLAB And Java.You will find libraries ready in MATLAB and Java languages to deal with Arduino, you can also use VB.NET with the different Arduino boards.

Types of Arduino boards

- ***ArduinoUNO***
- ***Arduino Mega***
- ***Arduino Nano***
- ***Arduino Mini***
- ***Arduino Lilypad***
- ***Arduino Demulive***
- ***Boarduino***



- The boards differ from each other in terms of the number of outputs and inputs, which determine the number of devices that can be controlled and the number of sensors that can be combined with the board, as well as the type of microcontroller and the speed of the processor located within and the possibility of switching or not.

ARDUINO UNO

- A small electronic circuit is used in a microcontroller of a company called ATMega328
- This circuit provides ports to connect electronic components directly to the controller via 14 digital in / out.
- Of these 14, there are 6 that can be used as PWM outputs (Pulse-Width modulation) and we will talk about this feature in a whole chapter called Analog Inputs & Outputs.

- The circuit also contains a Crystal Oscillator with 16MHz, plus a USB port for communication with the computer.
- There is a separate power input, in addition to the ICSP head, which is an additional method for programming the microcontroller while Connected to the board (other than USB).

Power up

- You can supply the circuit either through the USB port only or by using an external source of power as an AC / DC adapter to provide the circuit with the voltage required to operate or even by a 9 volt battery or 4 x 1.5 volt batteries where the two terminals are connected to the ground entrance GND and Vin in the circuit.

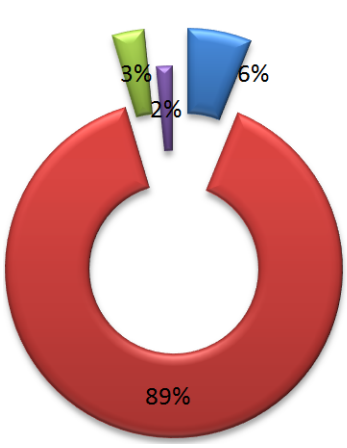
Power Inputs/Outputs

- Vin: Input voltage When we use an external power source, we can supply the voltage through this input, if we supply the power of the circuit through the input of the converter we can access it through this port as well.

- 5V: Regular voltage is used to supply the energy of the elements used on the circuit and we will use it to supply the energy of the electronic parts that we will add. This voltage may come through Vin via an internal voltage regulator or supplied through the USB port or any regular voltage source of 5 volts.
- 3.3V: Voltage source of 3.3 volts is guaranteed by the internal voltage regulator of the circuit and the maximum value for pulling the current through this line is 50 mA.
- GND: Landline

Microprocessor

- Microcontroller is like a small computer
- The ATmega328 microcontroller has a 16 MHz processor
- And the total memory is equal to 32Kilo Byte



ATMEGA 328P- MEMORY

- SRAM - 2 Killobyte
- Flash Disk - 29 Killobyte
- EEPROM - 1 Killobyte
- Bootloader 0.5 Killobyte

- Boot loader: The software is responsible for how to understand the circuits of the ArduinoC.
- SRAM: Memory used for storing variables temporary.
- Flash Disk: A storage space used to store the program that we will write to run the controller, this may seem very small size but in fact enough to write a lot of commands as we will see in the examples below.

- EEPROM: Memory is responsible for storing some variables permanently in the microcontroller and remain stored even after disconnecting the electricity and we can consider it as the Hard Disk in the personal computer.

Input & Output Pins(I/O)

- The 14 Digital Pins can be assigned as ports or outputs using program commands, and these lines operate at a maximum voltage of 5 volts and each line can provide a current of up to 40 mA, and there are 6 analogue input lines, A0 to A5, by default these inputs can measure voltage from 0 to 5 volts.